

ADAPTIVE DIGITAL SAT PRACTICE TEST

SAT Practice Test 3 - Standardized Edition

Section 1: Reading and Writing

Module 1 (27 questions) | Module 2 Easier (27) | Module 2 Harder (27)

Section 2: Math

Module 1 (22 questions) | Module 2 Easier (22) | Module 2 Harder (22)

Total: 147 questions

Section 1: Reading and Writing

Module 1 (27 Questions) - All Students

Time: 32 minutes

Question 1 Skill: Words in Context | Difficulty: Medium

The following text is from Kazuo Ishiguro's 2005 novel *Never Let Me Go*. Kathy, a young woman, is walking along a coastal path in Norfolk with her friend Tommy.

She watches Tommy **hurl** a stone across the choppy water, so far that it becomes a tiny dark speck before vanishing beneath the grey waves. The wind pulls at their jackets, stinging their cheeks red.

As used in the text, what does the word "hurl" most nearly mean?

- A) Strengthen
 - B) Shape
 - C) Celebrate
 - D) Throw
-

Question 2 Skill: Words in Context | Difficulty: Medium

Architect Maya Lin deeply admired the minimalist approach of Japanese garden design. Lin's memorial at the Vietnam Veterans site contains a long granite surface in tribute to that tradition that is a perfect _____ of Japanese simplicity: anyone familiar with those gardens would recognize the influence.

Which choice completes the text with the most logical and precise word or phrase?

- A) echo
 - B) dismissal
 - C) critique
 - D) theory
-

Question 3 Skill: Words in Context | Difficulty: Medium

The Singapore Exchange in Southeast Asia is a dedicated marketplace for investing in companies operating across the region, ensuring those companies meet certain transparency standards. Knowing this can _____ potential investors' concerns about regulatory gaps

and thereby allow them to instead focus on evaluating profitable ventures.

Which choice completes the text with the most logical and precise word or phrase?

- A) alleviate
 - B) broadcast
 - C) intensify
 - D) specify
-

Question 4 Skill: Words in Context | Difficulty: Medium

The habit of classifying musicians into rigid genre categories often encourages music critics to _____ subtleties in a performer's technique. Those analyzing the compositions of Duke Ellington, for instance, may inadvertently miss nuances in his work by focusing only on the most prominent ways in which his style corresponds to the swing era.

Which choice completes the text with the most logical and precise word or phrase?

- A) condemn
 - B) overlook
 - C) exaggerate
 - D) investigate
-

Question 5 Skill: Main Purpose | Difficulty: Medium

The following text is adapted from Chimamanda Ngozi Adichie's 2013 novel *Americanah*. Ifemelu is circulating a community survey among neighbors in her Lagos apartment building.

Ifemelu guarded the survey like a prized heirloom. She carried it with her at all times. In the building lobby, her gaze sharpened when she spotted a neighbor who hadn't yet responded. Wherever they were—at the mailboxes, near the elevator, in the parking garage or outside the laundry room—Ifemelu approached them. If they were collecting a delivery, she'd have them fill it out. If they were chatting, if they were rushing, if they were walking a dog home from the park, they answered her questions.

Which choice best states the main purpose of the text?

- A) To demonstrate that most neighbors are enthusiastic about completing the survey
- B) To show that attitudes toward the survey within the building change over time

- C) To portray Ifemelu’s strong commitment to collecting responses for the survey
- D) To suggest that some neighbors refuse to complete the survey because they dislike Ifemelu

Question 6 Skill: Function (Underlined Portion) | Difficulty: Medium

In Quechua, an Indigenous language family spoken across the Andes region of South America, *wasi* means “house,” whereas *wasi-wasi* means “village” or “many houses.” This phenomenon, in which an element of a root word is repeated, sometimes with modification, within another word that is related to the root word, is called reduplication. In this case, the full word *wasi* is repeated to form *wasi-wasi*. There are many examples of this type of reduplication in Quechua.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A) It provides a general definition of the phenomenon discussed in the text.
- B) It explains where the language discussed in the text is from.
- C) It contrasts the language discussed in the text with other languages.
- D) It provides English translations of the Quechua words discussed in the text.

Question 7 Skill: Detail (According to the Text) | Difficulty: Medium

Nanoplastics are extremely small fragments of plastic debris. Regions of rivers with higher concentrations of nanoplastic particles also exhibit reduced current velocity. A study by Haruna Yamamoto and colleagues found that the concentration of nanoplastic particles cannot be the sole explanation for this reduced flow because the concentration of particles that would produce the observed effect is much higher than that found in these sections of the river. However, they found that emulsifiers, chemicals frequently used in plastic manufacturing, are released into the water from nanoplastics and have a much stronger flow-reducing effect.

According to the text, what did Yamamoto and colleagues discover about emulsifiers?

- A) They can be used to contain nanoplastics within certain areas of rivers.
- B) They are helpful for filtering nanoplastics from river water.
- C) They have a much stronger effect on water flow than nanoplastics alone do.

D) They are mainly composed of dissolved minerals.

Question 8 Skill: Main Idea | Difficulty: Medium

Synthetic photovoltaic cells are an emerging clean energy technology that replicates the process of photosynthesis found in plants. These devices are polymer-based panels coated in metallic catalysts that drive reactions splitting carbon dioxide into carbon monoxide and oxygen. The technology, while generating significant academic interest, is not yet commercially feasible as a large-scale power source. To address this challenge, engineers from several disciplines are researching ways to store, transport, and distribute the energy the devices produce while other engineers are working to improve the cost and efficiency of the devices.

Which choice best states the main idea of the text?

- A) Continued research and development in synthetic photovoltaic technology is needed before the devices can be widely used as an energy source.
- B) Synthetic photovoltaic cells were developed to replicate the natural process of photosynthesis in order to store energy for long-term commercial use.
- C) The recent increase in the commercial use of synthetic photovoltaic cells as an energy source has encouraged many engineers to research ways to improve the technology.
- D) Synthetic photovoltaic cells split carbon dioxide into carbon monoxide and oxygen using catalysts more efficiently than plants do using photosynthesis.

Question 9 Skill: Rhetorical Synthesis | Difficulty: Medium

[Table: Total Areas of Five Indigenous Territories in Canada]

Indigenous territory	Province/Territory	Area (square kilometers)
Nisga’a Lands	British Columbia	2,019
Nunavik	Quebec	660,000
Labrador Inuit Settlement Area	Newfoundland and Labrador	72,520
Sahtu Settlement Area	Northwest Territories	41,437
Tłıchǫ Lands	Northwest Territories	39,000

A member of the Tłıchǫ Nation is a student in a geography class. The student wants to compare the total area covered by his people’s territory in the Northwest Territories with the total areas covered by cer-

tain Indigenous territories in other provinces. Looking at the table, he finds that the area of Tłıchq Lands is 39,000 square kilometers, while the area of the Nisga'a Lands in British Columbia is

Which choice most effectively uses data from the table to complete the comparison?

- A) 660,000 square kilometers.
- B) 72,520 square kilometers.
- C) 2,019 square kilometers.
- D) 41,437 square kilometers.

Question 10 Skill: Evidence - Data / Graph / Table | Difficulty: Medium

GRAPH TO INSERT

[Table: Numbers of 18 Non-native Shrub Species Reported and the Insect and Fungus Threats to Them]

Country	Shrubs	Fungi	Insects
Italy	8	19	87
Greece	3	9	7
Portugal	5	14	22

Katarina Voss and colleagues gathered data on 18 non-native shrub species grown in southern Europe. They analyzed reports from Italy, Greece, and Portugal about the number of these species grown in those countries as well as the numbers of insect and fungus species that damage those shrubs. The researchers concluded that Italy reported a greater number of damaging insect species than either of the other countries did.

Which choice best describes data from the table that support Voss and colleagues' conclusion?

- A) Greece reported 9 damaging fungus species and 7 damaging insect species, while both Italy and Portugal reported more insect species than fungus species.
- B) Italy reported 19 damaging fungus species, which is more than either Greece or Portugal reported.
- C) Portugal and Greece reported 5 and 3 damaging shrub species, respectively, which is far fewer than Italy reported.
- D) Italy reported 87 damaging insect species, which is more than either Portugal or Greece reported.

Question 11 Skill: Evidence - Support / Weaken | Difficulty: Medium

Priya Venkatesh, Anil Gupta, and colleagues assembled a database of compounds used in cosmetic products by testing formulations in the lab and by gathering data from other similar product studies. The team rated each study on its level of specificity (with a higher rating for reported data with more specificity) and on its classification alignment (with a higher rating for using compound categories more closely aligned with the categories in the team’s database). Based on these ratings, a second research team concluded that the data in a study by David Chen and colleagues would need less adjustment to fit the classifications in the database than the data in a study by Laura Schmidt and colleagues did.

Which finding, if true, would most directly support the second research team’s conclusion?

- A) The study by Chen and colleagues had a higher alignment rating than the study by Schmidt and colleagues did.
- B) The study by Chen and colleagues had a high alignment rating and a medium specificity rating.
- C) The study by Chen and colleagues had a higher specificity rating than the study by Schmidt and colleagues did.
- D) The study by Schmidt and colleagues had a medium alignment rating and a medium specificity rating.

Question 12 Skill: Evidence - Data / Graph / Table | Difficulty: Medium

GRAPH TO INSERT

[Table: Ranking of Environmental and Social Benefits of Community Gardens (scale of 1 to 20; 1 = highest)]

Social or environmental benefit	Garden managers	Volunteers	Local residents
mental health improvement	7	12	10
reduction of food waste	11	9	18
supply of fresh vegetables	19	17	3
habitat for pollinators	4	6	14
beautification of public spaces	14	18	16

Tomás Reyes, Ingrid Lund, and their team surveyed three groups of

people in Portland, Oregon—managers of community garden projects, volunteers in community gardens, and local residents—to compare their views about the extent to which community gardens contribute to 20 social or environmental benefits that the team identified. The researchers used these ratings to rank the benefits for each group, with a ranking of 1 indicating that a group perceives community gardens as contributing that benefit the most. Using only the rankings shown in the table, a city official who is promoting a new community garden project concludes that advertisements aimed at local residents should emphasize the project’s benefit of supplying fresh vegetables.

Which choice best describes data in the table that support the city official’s conclusion?

- A) The reduction of food waste was ranked higher for local residents than was the supply of fresh vegetables.
- B) The supply of fresh vegetables was ranked higher for volunteers than it was for local residents.
- C) The mental health improvement benefit was ranked lower for local residents than it was for garden managers.
- D) The supply of fresh vegetables was ranked higher for local residents than were the other four benefits.

Question 13 Skill: Evidence - Support / Weaken | Difficulty: Medium

Marina Ospina, Kenji Tanaka, and Fatima Al-Rashid investigated the evolution and function of bioluminescence in deep-sea shrimp and krill species. They found that during harmless simulated encounters with predators on isolated specimens, 28% of the tested species produced light, which ranged from steady glows in *Oplophorus gracilirostris* to rapid flashes in *Meganyctiphanes norvegica*. Although some marine organisms use bioluminescence to communicate with members of the same species, the researchers claim that the light signals recorded in their study are directed primarily at predators.

Which finding, if true, would most directly support Ospina and colleagues’ claim?

- A) None of the species that produced light did so before the simulated encounters, and nearly all stopped producing light within ten seconds after the encounters.
- B) Shrimp glows were emitted in a wavelength detectable by fish that prey on shrimp, but krill flashes were not.
- C) Each species tended to produce one type of light signal during simulated encounters, although individuals occasionally produced a variety of other signals during simulated encounters as well.

- D) In most cases, the light signal that a species produced during simulated encounters was not produced by other species during simulated encounters.

Question 14 Skill: Inference / Complete the Text | Difficulty: Medium

Over 300 languages are spoken in London in addition to English—one can find Somali spoken in the borough of Tower Hamlets, or Tamil in Harrow. Many immigrants from northern India, where Hindi is the dominant language, reside in the borough of Southall (part of the London Borough of Ealing) while those who emigrated from the Sylhet region of Bangladesh, where many people speak Sylheti as a first language, tend to reside in the borough of Tower Hamlets. Among speakers of South Asian languages, those in Southall tend to speak Hindi as their primary language while those in Tower Hamlets tend to speak Sylheti or Bengali as their primary language. This is most likely because

Which choice most logically completes the text?

- A) South Asian immigrants who emigrated to London many years ago are more likely to speak several languages than are more recent immigrants.
- B) taken together, there are more Sylheti and Bengali speakers among South Asian immigrants in London than there are Hindi speakers.
- C) South Asian immigrants regularly change their residences between Ealing and Tower Hamlets after they emigrate, rather than staying in one borough.
- D) there are geographic differences in primary language within South Asia that are reflected in the settlement patterns of South Asian immigrants in London.

Question 15 Skill: Grammar / Conventions | Difficulty: Medium

Florence Nightingale's statistical diagrams were printed in government reports in 1858, making the pioneering nurse one of many healthcare professionals _____ on public health documents.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) did contribute
- B) had contributed
- C) contributes
- D) to appear

Question 16 Skill: Grammar / Conventions | Difficulty: Medium
(*restrictive Clause*)

Several urban planners, like transportation analyst Roberto Diaz-Fernandez, have proposed expanding light rail networks as one possible way to reduce traffic on the congested highways of Harris County, Texas. Indeed, local transit routes like the METRORail Purple Line _____ have become an increasingly popular means of travel for commuter and leisure trips alike.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) route, which is 6 miles long
- B) route which is 6 miles long—
- C) route, which is 6 miles long,
- D) route: which is 6 miles long.

Question 17 Skill: Grammar / Conventions | Difficulty: Medium
(*participle*)

Montana, _____ 4,506 feet in elevation, reaches its highest point at Granite Peak.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) spanning
- B) spans
- C) span
- D) spanned

Question 18 Skill: Grammar / Conventions | Difficulty: Medium
(*positive*)

The International Monetary Fund (IMF) tracks comparative purchasing power data for its member countries. According to this data, in March 2022, a “basket” of goods and services priced at 100 US dollars (USD) in the United States would have cost 112 USD in fellow IMF member _____

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) nation, Norway.
- B) nation—Norway.
- C) nation Norway.

D) nation: Norway.

Question 19 Skill: Grammar / Conventions | Difficulty: Medium

While the Bengal tiger can be found in countries like Nepal and Bhutan, more than 70 percent of this endangered cat species is found in India. There, wildlife conservationist Dr. Raghu Chundawat is on the front lines of conservation efforts _____ through habitat restoration and scientific research, aim to bring tigers back from near extinction.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) that—
 - B) that;
 - C) that,
 - D) that:
-

Question 20 Skill: Grammar / Conventions | Difficulty: Medium
(Agreement)

In graph theory (the mathematical study of networks and connections), graphs are characterized by their vertex counts—that is, the number of points where edges meet. The star graph and the cycle graph, each with a vertex count below six, _____ among the simplest possible graphs with the fewest vertices.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) are
 - B) has been
 - C) is
 - D) was
-

Question 21 Skill: Grammar / Conventions | Difficulty: Medium
(positive)

At over 2,000 years old, the Fortingall Yew (*Taxus baccata*) located in Perthshire, Scotland, was one of the oldest known trees in Europe. With almost two millennia of growth data in its trunk rings, a single tree like this, claims _____ Valerie Trouet, can tell the history of the continent.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) dendrochronologist,
 - B) dendrochronologist—
 - C) dendrochronologist
 - D) dendrochronologist:
-

Question 22 Skill: Transitions | Difficulty: Medium

Marine biologist Sylvia Earle is enthusiastic about the capabilities of the new Alvin deep-sea submersible. Unlike its predecessor, the older Johnson-Sea-Link, the Alvin can withstand pressures at depths exceeding 6,000 meters. Most hydrothermal vents are found at extreme depths; _____ the Alvin can provide Earle and her team much more data about these vents than the Johnson-Sea-Link could.

Which choice completes the text with the most logical transition?

- A) in contrast,
 - B) consequently,
 - C) additionally,
 - D) actually,
-

Question 23 Skill: Transitions | Difficulty: Medium

During many historic London celebrations, including the 1953 coronation procession for Queen Elizabeth II, the colorful confetti drifting through the streets were small paper discs cut from ledger rolls, accounting documents that were used to record financial transactions. _____ the ledger rolls had long since been replaced by computers by the time of the celebration for the 2012 Olympics. Then, commercially manufactured confetti brightened the festivities.

Which choice completes the text with the most logical transition?

- A) Of course,
 - B) For example,
 - C) In addition,
 - D) As a result,
-

Question 24 Skill: Rhetorical Synthesis | Difficulty: Medium

- *Reptiles of the Caribbean Basin* is an identification guidebook by herpetologists Marco Salinas, Devon Archbold, and Lucia Ferreira.
- It lists the twenty-eight gecko species found in Puerto Rico.
- The crested gecko is a medium-sized gecko found in Puerto Rico.

- It is identifiable by its distinctive ridged crest and its long, prehensile tail.
- The mourning gecko is a small gecko found in Puerto Rico.
- It is identifiable by its dark dorsal stripe and its short, rounded snout.

Which choice most effectively uses information from the given sentences to emphasize a difference between the two geckos?

- A) One way to distinguish the crested gecko from the mourning gecko is to look at their tails: the crested gecko's is long and prehensile, whereas the mourning gecko's is not described as such.
- B) Identifiable by its long, prehensile tail and its distinctive ridged crest, the crested gecko is a medium-sized gecko found in Puerto Rico.
- C) The mourning gecko is a small gecko identifiable by its dark dorsal stripe and its short, rounded snout.
- D) The crested gecko and the mourning gecko are two of the twenty-eight different gecko species found in Puerto Rico.

Question 25 *Skill: Rhetorical Synthesis | Difficulty: Medium*

While researching a topic, a student has taken the following notes:

- The Julian calendar was introduced by Julius Caesar in 46 BCE.
- The Gregorian calendar replaced the Julian calendar in most of Europe starting in 1582.
- Both calendars are divided into twelve months.
- The lengths of certain months differ between the two calendars.
- The Julian calendar assigns 28 days to February in common years, while the Gregorian calendar also assigns 28 days to February but adjusts leap year rules.
- The Julian calendar's leap year rule adds a day every four years without exception, beginning in March.

The student wants to emphasize a difference between the Julian and Gregorian calendars. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Introduced by Julius Caesar in 46 BCE, the Julian calendar includes twelve months and was eventually replaced by the Gregorian calendar in 1582.
- B) The two calendars handle leap years differently; for example, the Julian calendar adds a leap day every four years without exception, while the Gregorian calendar introduced additional rules to improve accuracy.

- C) The Gregorian calendar, which replaced the Julian calendar in most of Europe starting in 1582, has twelve months, including February.
 - D) The twelve months of the Julian calendar were established by Julius Caesar in 46 BCE.
-

Question 26 *Skill: Rhetorical Synthesis | Difficulty: Medium*

While researching a topic, a student has taken the following notes:

- The Beaufort scale of wind force is a system that classifies wind speeds by their observed effects.
- It was originally designed for use at sea by Sir Francis Beaufort in 1805.
- Wind speeds are ranked from 0 (calm) to 12 (hurricane force).
- At Beaufort number 3 (gentle breeze), wind speed is 7–10 knots.
- At Beaufort number 7 (near gale), wind speed is 28–33 knots.
- At Beaufort number 10 (storm), wind speed is 48–55 knots.

The student wants to compare the intensity of three wind categories. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Based on their Beaufort numbers, a storm (number 10) involves stronger winds than a near gale (number 7), and a near gale involves stronger winds than a gentle breeze (number 3).
 - B) The Beaufort scale of wind force can be used to classify gentle breezes, near gales, and storms by their observed effects on the environment.
 - C) A wind speed of 48–55 knots, classified as a storm, is assigned a higher Beaufort number than a gentle breeze, which has a speed of 7–10 knots.
 - D) Storms can produce visible effects on the sea surface, which is why Sir Francis Beaufort originally designed the scale for maritime use in 1805.
-

Question 27 *Skill: Rhetorical Synthesis | Difficulty: Medium*

- Dolores Huerta was a Mexican American labor leader and civil rights activist.
- Between 1962 and 2000, she co-founded and led several farm-worker advocacy organizations.
- She co-founded the National Farmworkers Association in Delano, California, in September 1962.
- She helped organize the Delano grape strike in Kern County, California, in 1965.

- Her advocacy efforts are archived at the Walter P. Reuther Library.
- The archives can now be accessed online through the Reuther Library's digital collection.

Which choice most effectively uses information from the given sentences to begin a narration of Huerta's co-founding of the National Farmworkers Association?

- A) While both the National Farmworkers Association and the Delano grape strike involved Dolores Huerta's leadership, the Association was co-founded in Delano, California, and the strike took place in Kern County, California.
 - B) It was in September 1962, in the agricultural town of Delano, California, that Dolores Huerta co-founded a new organization to champion the rights of farmworkers: the National Farmworkers Association.
 - C) Decades of advocacy materials collected by Dolores Huerta can now be found in one place: online at the Walter P. Reuther Library's digital collection.
 - D) Dolores Huerta co-founded the National Farmworkers Association in September of 1962 and helped organize the Delano grape strike in 1965.
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Section 1: Reading and Writing

Module 2 - Easier Path (27 Questions)

Time: 32 minutes

Question 1 *Skill: Words in Context | Difficulty: Easy*

Sociologist Marta Reyes argues that the consequences of global migration may display regional variation at local levels. For instance, remittances may boost a community's average income, but for families in remote highland villages, steep internal transportation costs could _____ the financial benefits associated with those funds.

Which choice completes the text with the most logical and precise word or phrase?

- A) underestimate
 - B) misconstrue
 - C) denigrate
 - D) nullify
-

Question 2 *Skill: Words in Context | Difficulty: Easy*

As discussed by scholar Tomoko Ishikawa, many interactive web-based art installations produced in the late 1990s to the early 2010s exclusively for exhibition on the internet, such as Olia Lialina's *My Boyfriend Came Back from the War* (1996), have become inaccessible because viewing them requires the use of _____ software (most notably Java applets, largely unsupported since 2017).

Which choice completes the text with the most logical and precise word or phrase?

- A) extraneous
- B) arcane
- C) defunct
- D) ubiquitous

Question 3 Skill: Words in Context | Difficulty: Easy

Proposals to lower the threshold at which small businesses begin paying corporate taxes are generally discussed in terms of the effects on government revenue, but economist Andrea Fiorini has argued that implementing such changes could _____ economic growth among entrepreneurs by increasing the administrative burden during the period in which they are scaling their operations and seeking investors.

Which choice completes the text with the most logical and precise word or phrase?

- A) disparage
- B) outstrip
- C) compound
- D) stymie

Question 4 Skill: Words in Context | Difficulty: Easy

In the late 2000s, the value of vintage vinyl records surged dramatically, which had the counterintuitive effect of _____ demand: collectors who hadn't previously sought out rare pressings flooded the market, believing prices would continue to climb and the records could be resold later at a profit.

Which choice completes the text with the most logical and precise word or phrase?

- A) exploiting
- B) engendering
- C) monetizing
- D) appraising

Question 5 Skill: Overall Structure | Difficulty: Easy

In 2014, Keiko Yamamoto and colleagues published a study concluding that rising ocean temperatures have a strong effect on the migration patterns of *Thunnus alalunga*, a species of tuna. However, Yamamoto and colleagues' study relied on a tracking sample of only about 30 fish. In a 2023 review of various scientists' conclusions about the impacts of ocean warming on fish migration, Carlos Mendez and colleagues caution that relying on such a small sample size can increase the potential for biased analysis. Such analysis, in turn, can contribute to reports of exaggerated effects.

Which choice best describes the overall structure of the text?

- A) It states a similarity between two scientific studies, then notes a difference between them.
- B) It describes a characteristic of a fish species, then explains why that characteristic is noteworthy.
- C) It presents the result of a study, then raises a potential concern related to that result.
- D) It summarizes a problem that scientists are investigating, then provides a possible solution to that problem.

Question 6 Skill: Function (Underlined Portion) | Difficulty: Easy

Researchers Lin Wei, Dmitri Petrov, and Amara Osei created a computer model to predict the mix of restaurants, retail shops, and other businesses found in a given urban district. How we define a district and its boundaries is subjective, so the team used a clustering algorithm to locate dense groupings of commercial activity that represent organically defined neighborhoods like San Francisco's Mission District. The predictive model, which incorporates this algorithm, is sure to be invaluable in determining the optimal mix of a district's businesses.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A) It praises an algorithm's accuracy in identifying district boundaries.
- B) It emphasizes the potential utility of the team's model.
- C) It summarizes trends in recent commercial development in San Francisco.
- D) It suggests a difficulty associated with analyzing districts that the research team attempted to overcome.

Question 7 Skill: Main Purpose | Difficulty: Easy

Marco Alvarez and colleagues have explored how parallel evolution—a phenomenon that occurs when similar traits develop independently in two closely related but geographically separated lineages—can result from shared environmental pressures acting on both lineages. Meanwhile, Priya Sharma and colleagues have investigated how parallel evolution occurs through different adaptive pathways, but the relative frequency of parallel evolution through shared and different adaptive processes is still poorly understood. This motivated biologists Yuki Tanaka and Samuel Okonkwo to evaluate both types of parallelism in a single study for their 2019 paper.

Which choice best states the main purpose of the text?

- A) To suggest that Tanaka and Okonkwo’s conclusions would be more persuasive if they incorporated findings from the studies by Alvarez and colleagues and Sharma and colleagues
- B) To argue that Tanaka and Okonkwo’s study revealed a previously unrecognized issue that evolutionary biologists need to address
- C) To summarize Tanaka and Okonkwo’s criticism of the study by Alvarez and colleagues
- D) To situate Tanaka and Okonkwo’s study in the context of their field

Question 8 Skill: Cross-Text | Difficulty: Easy

Text 1

Literary criticism today disproportionately emphasizes autobiographical and confessional narrative modes, such as that of Sylvia Plath’s *The Bell Jar*, beyond the degree to which they actually influenced fiction in Britain during the postwar period (roughly 1945–1980). Meanwhile, Muriel Spark’s *The Prime of Miss Jean Brodie*, whose satirical, detached narrative voice recalls the fiction of earlier decades, attracts remarkably little attention from scholars of postwar literature.

Text 2

Computational stylistics, or algorithm-assisted quantitative analysis of large collections of digitized literary texts, can reveal tonal and structural elements that have previously escaped notice, despite being shared by numerous texts from the postwar period. For too long, scholars have focused on confessional sincerity versus ironic detachment, inhibiting inquiry into other points of stylistic correspondence among works that would enrich our understanding of the postwar canon.

Based on the texts, both the author of Text 1 and the author of Text 2 would most likely agree with which statement about scholarship on the postwar period in Britain?

- A) Its primary methods for analyzing fiction written in the period are growing obsolete as computer technology advances.
- B) Instead of engaging in unproductive debates, it should work to rehabilitate the reputations of neglected postwar works.
- C) At present, it only partially captures the stylistic dimensions of the fiction written in the period.
- D) It must widen its focus to include aspects of postwar fiction beyond style, a productive but overrepresented site of inquiry.

Question 9 Skill: Main Idea | Difficulty: Easy

Why do some people with modest incomes vote for politicians supporting expanded social programs even when those programs are funded by taxes on lower-income brackets? Behavioral economists Sarah Lindqvist et al. propose that community solidarity is a practical consideration: that is, the more deeply embedded one is in a close-knit community, the more one feels obligated to support communal welfare through one's vote. Thus, Lindqvist et al. suggest that although the behavior of lower-income voters who support higher taxes on themselves may seem counterintuitive, such people likely do so because they feel motivated by their social bonds to take a stance they believe benefits their neighbors.

Which choice best states the main idea of the text?

- A) A group of economists asserts that people with strong community ties are consequently motivated to take certain considerations into account when voting.
- B) A team of economists finds that people who vote for higher taxes on lower incomes are likely to think their communal values coincide with their material interests.
- C) According to a group of economists, politicians who support expanded social programs must convince a sufficient number of modest-income voters to vote against their immediate economic interest if the politicians are to be elected.
- D) According to a team of economists, the more embedded a voter is in a community, the more likely that voter's financial interests conflict with their social obligations.

Question 10 Skill: Evidence - Data / Graph / Table | Difficulty: Easy

GRAPH TO INSERT

[Table: Studies of Gray Wolf Population Density]

Study authors	Study publication year	Location	Minimum density (wolves per 1,000 square kilometers)	Maximum density (wolves per 1,000 square kilometers)
Marco Ciucci et al.	2015	Abruzzo (Italy)	4.20	7.80
Olof Liberg et al.	2012	Scandinavia (Sweden/Norway)	0.95	2.10
Luigi Boitani et al.	2018	Tuscany (Italy)	2.30	3.50
Josip Kusak et al.	2011	Gorski Kotar (Croatia)	2.80	6.40

Studies of the population density of gray wolves (*Canis lupus*) have yielded a range of results, which may in part reflect natural variations in the resources that wolves need to survive. For example, the difference between the maximum population density reported by Olof Liberg et al. and that reported by Marco Ciucci et al. may indicate that

Which choice most effectively uses data from the table to complete the example?

- A) the wolf habitat in Tuscany can support more than 2.30 individuals per 1,000 square kilometers.
- B) the wolf habitat in Scandinavia cannot support as many individuals as can the wolf habitat in Abruzzo.
- C) wolf habitat makes up a greater proportion of the overall land area in Italy than is the case in Scandinavia.
- D) there are more wolves overall in both Tuscany and Croatia than there are in Scandinavia.

Question 11 Skill: Quotation Illustrates Claim | Difficulty: Easy

The House of Mirth is a 1905 novel by Edith Wharton. In the novel, the narrator portrays the character Lily Bart as someone who admires wealth in other people to the exclusion of all other qualities: _____

Which quotation from *The House of Mirth* most effectively illustrates the claim?

- A) "Indeed from the very beginning Lily showed her preference for luxury."
 - B) "Lily was like her mother not only so far as social ambition was concerned but also in her desire for comfort."
 - C) "Lily was immensely interested in people who possessed great fortunes. Otherwise not at all."
 - D) "She had a natural grace, an instinctive taste for elegance, and a remarkably vivid and expressive face."
-

Question 12 Skill: Evidence - Support / Weaken | Difficulty: Easy

Some industrial solvents contain volatile organic compounds (VOCs), which can evaporate into indoor air and accumulate in building ventilation systems. In a 2018 study, Rina Patel and Jae-Won Kim found that VOCs can concentrate in the filters of standard HVAC systems (*Carrier 58MCA*). While accumulation of industrial chemicals may be inherently concerning, it has been hypothesized that VOC accumulation in HVAC filters like those in *Carrier 58MCA* systems could serve a valuable detection role, obviating the need for building managers to conduct costly and invasive air-quality sampling using specialized equipment—such as photoionization detectors (PIDs), commonly required in regulatory compliance testing—for VOC monitoring, as workplace safety laws currently require.

Which finding, if true, would most directly support the hypothesis presented in the text?

- A) In comparable buildings, *Carrier 58MCA* systems and PIDs display comparable rates of VOC detection.
 - B) Compared with PIDs, *Carrier 58MCA* filters can tolerate significantly higher VOC concentrations without requiring replacement.
 - C) It is easier to detect low and harmless concentrations of VOCs in *Carrier 58MCA* filters than it is to detect high and harmful concentrations of VOCs using PIDs.
 - D) VOC concentrations in *Carrier 58MCA* filters tend to vary more from unit to unit than do VOC readings from PIDs when the systems are exposed to similar levels of VOCs.
-

Question 13 Skill: Evidence - Support / Weaken | Difficulty: Easy

Neuroscientists Akira Watanabe, Elena Bortolotti, and colleagues investigated the scent detection abilities of twelve domestic cats of various ages. The researchers monitored the brain activity of Mochi (a 96-month-old Siamese), Ziggy (a 108-month-old tabby), and other cats while the animals were exposed to three scent samples: one of fish

extract, the second of catnip, and a third made up of randomly combined synthetic molecules scrambled so that they didn't resemble any natural scent. Each of the cats was familiar with either fish or catnip, but not both. The team concluded that the younger the cat, the worse it may be at differentiating between familiar and unfamiliar scents.

Which finding from the study, if true, would most directly support the team's conclusion?

- A) As the age of the cat scanned decreased, so did the amount of brain activity in response to smelling the scent the cat was accustomed to or the other scent, but not in response to the scrambled sample.
- B) The similarity between the pattern of brain activity a cat showed in response to the scrambled sample and the pattern it showed in response to the scent it was not accustomed to was lowest among younger cats.
- C) Cats showed a different pattern of brain activity when smelling the scent they were accustomed to than when smelling the other scent, and the difference in brain activity decreased as the age of the cat scanned decreased.
- D) Although the cats' general olfactory sensitivity declined with age, cats of all ages showed more brain activity in response to smelling the scent they were accustomed to than in response to smelling the other scent.

Question 14 Skill: Quotation Illustrates Claim | Difficulty: Easy

Songs of the Open Road is a 1912 collection of verse by Madison Cawein. In one of his poems, the speaker indicates that several new freedoms and possibilities have recently become available to the working class, saying, _____

Which quotation from *Songs of the Open Road* most effectively illustrates the claim?

- A) "Simple child of sun and rain, / [The worker] toiled beneath the open sky." (from "The Laborer")
- B) "We must take the world's rough road, / All its burdens we shall bear; / Dawn was meant for honest toil, / Dusk was meant for rest." (from "The Worker's Hymn")
- C) "For horizons [the worker] had not dared to seek, / Till hardship taught how wide the world could be." (from "The Laborer")
- D) "Mill and furnace, forge and field, / Have opened wider paths before the common hand; / And greater grows the worker's honest yield, / More justly earned, and more humanely planned." (from "The Laborer")

Question 15 Skill: Inference / Complete the Text | Difficulty: Easy

Pulsars—such as the Vela Pulsar, located in the Vela constellation—are rapidly rotating neutron stars that emit beams of electromagnetic radiation. Pulsars vary in rotational period, with approximately 30 of them known to spin faster than 100 times per second. Astrophysicists have long wondered how any pulsars could rotate so rapidly given that conditions during neutron star formation are believed to have been ill-suited to producing such extreme spin rates, which suggests that _____

Which choice most logically completes the text?

- A) the Vela Pulsar is likely slower than pulsars that formed more than a billion years after the first neutron stars.
- B) pulsars that formed in the early universe are likely not as fast as those that formed later.
- C) the Vela Pulsar is thought to have formed less than a billion years after the first neutron stars appeared.
- D) some aspect of the scientific understanding of pulsar formation or neutron star evolution may be incomplete.

Question 16 Skill: Inference / Complete the Text | Difficulty: Easy

To avoid detection by echolocating dolphins and other marine predators, many squid species, including *Dosidicus gigas*, change skin color rapidly through chromatophore cells that, in some cases, confuse the sonar signals dolphins rely on for hunting. Some marine biologists have hypothesized that this capability evolved because it imposes a lower metabolic cost than does the alternative defense mechanism of producing ink clouds that obscure the squid from predators. Alejandra Muñoz et al. investigated the optical properties of squid chromatophore displays in response to recorded dolphin echolocation and then assessed the toxicity of the ink produced by color-changing species. They found that several squid genera whose chromatophore displays are capable of disrupting dolphin echolocation also produce highly toxic ink, suggesting that _____

Which choice most logically completes the text?

- A) further investigations into squid defense mechanisms will likely find that genera relying on chromatophore displays are also generally inedible to dolphins.
- B) most genera of squid that produce displays capable of disrupting dolphin echolocation do so primarily for purposes other than evading capture by dolphins and other predators.

- C) the hypothesis about the development of this visual defense likely does not account for all instances of the trait in squid.
 - D) although previous findings about *Dosidicus gigas* are consistent with the hypothesis about the low metabolic cost of producing toxic ink, the ability to disrupt dolphin echolocation and ink toxicity are not mutually exclusive traits.
-

Question 17 Skill: Grammar / Conventions | Difficulty: Easy

Researchers Amira Hassan and Kenji Nakamura have lent their expertise on African American culinary history to an important new _____ Timeline of Soul Food, a digital resource that traces the development of individual regional cuisines (such as Lowcountry and Creole) while also revealing the connections between them.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) initiative the
 - B) initiative; the
 - C) initiative: the
 - D) initiative. The
-

Question 18 Skill: Grammar / Conventions | Difficulty: Easy
(positive)

The Bantu language family includes Swahili, Zulu, and some 500 other languages throughout sub-Saharan Africa, making it one of the largest language families in the world and of keen interest to _____ of the University of Cape Town.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) linguist, Nomsa Dlamini
 - B) linguist: Nomsa Dlamini
 - C) linguist, Nomsa Dlamini,
 - D) linguist Nomsa Dlamini
-

Question 19 Skill: Grammar / Conventions | Difficulty: Easy

Located on the Rhine River in Germany, the city of Cologne was a member of a powerful mercantile alliance that dominated northern European trade between the 12th and 16th _____ Hanseatic League, a loose confederation of cities from multiple modern-day countries that has been described as a precursor to today's European Union.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) centuries; the
 - B) centuries: the
 - C) centuries the
 - D) centuries. The
-

Question 20 Skill: Grammar / Conventions | Difficulty: Easy
(Agreement)

Kehinde Wiley is known for using portraiture and sculpture to explore questions of race and power in the representation and presentation of Black identity. The acclaimed artist was awarded a 2015 residency by the American Academy in Rome, whose extensive collection of artworks by European masters and contemporary artists of the Americas and the Mediterranean _____ among the finest in the world.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) have been ranked
 - B) rank
 - C) are ranked
 - D) ranks
-

Question 21 Skill: Grammar / Conventions | Difficulty: Easy
(positive)

Many of the world's most celebrated marine animals have lived in the Monterey Bay National Marine Sanctuary in California. Between 2015 and 2018, for instance, the _____ patrolled the sanctuary as part of the resident great white shark population.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) sharks Scarface and Luna,
 - B) sharks Scarface and Luna
 - C) sharks, Scarface and Luna,
 - D) sharks, Scarface and Luna
-

Question 22 Skill: Grammar / Conventions | Difficulty: Easy
(structure)

Award-winning actor Lupita Nyong'o, who has appeared in over 20 major feature films, is one of numerous luminaries to be photographed

by Kenyan portraitist Osborne Macharia for his exhibition *Nairobi Noir*. _____ referred to Kenya's thriving film scene as "East Africa's vibrant canvas par excellence," celebrates its legacy with his vivid stylized portraits of Nyong'o and her contemporaries.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) Macharia has
- B) Macharia
- C) Macharia, has
- D) Macharia, having

Question 23 Skill: Transitions | Difficulty: Easy

Newer solar panel arrays often outperform older installations. For example, Germany's Waldpolenz Solar Park, built in 2008, had panels that generated about 40 megawatts (MW) of electricity total. _____ Germany's Weesow-Willmersdorf Solar Park, completed in 2020, has panels that can generate about 187 MW of electricity total.

Which choice completes the text with the most logical transition?

- A) In contrast,
- B) Granted,
- C) Eventually,
- D) For this reason,

Question 24 Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

- Titanoboa is the common name of a fossil snake specimen from the Paleocene epoch.
- The Paleocene epoch ended more than 55 million years ago.
- Titanoboa is a member of the family Boidae.
- A reconstructed Titanoboa model is on display at the Smithsonian National Museum of Natural History.
- The Smithsonian National Museum of Natural History is in Washington, D.C.

The student wants to specify Titanoboa's location. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Washington, D.C., is home to the Smithsonian National Museum of Natural History, which displays a reconstructed model of a fossil snake from the Paleocene epoch.

- B) Titanoboa lived in the Paleocene epoch, which ended more than 55 million years ago.
 - C) A reconstructed model of Titanoboa is on display at the Smithsonian National Museum of Natural History in Washington, D.C.
 - D) Titanoboa is the common name of a fossil snake specimen belonging to the family Boidae.
-

Question 25 *Skill: Rhetorical Synthesis | Difficulty: Easy*

While researching a topic, a student has taken the following notes:

- The pH scale measures how acidic or basic a substance is.
- It ranges from 0 (most acidic) to 14 (most basic).
- A substance with a pH of 7 is considered neutral.
- Lemon juice has a pH of approximately 2.
- Baking soda dissolved in water has a pH of approximately 9.
- Household bleach has a pH of approximately 12.5.

The student wants to compare the acidity or basicity of three substances. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The pH scale ranges from 0 to 14, with 7 being neutral; lemon juice, baking soda solution, and bleach all fall at different points on this scale.
 - B) Lemon juice, with a pH of approximately 2, is highly acidic, while baking soda solution (pH 9) and bleach (pH 12.5) are both basic, with bleach being the most basic of the three.
 - C) Household bleach is a very basic substance with a pH of approximately 12.5, which is why it is an effective cleaning agent.
 - D) Baking soda dissolved in water has a pH of approximately 9, making it a mildly basic substance on the pH scale.
-

Question 26 *Skill: Rhetorical Synthesis | Difficulty: Easy*

- Carmen Miranda (1909–1955) was a performer born in Marco de Canaveses, Portugal.
- She was raised in Rio de Janeiro, Brazil.
- She was of Portuguese heritage.
- She was the first Latin American entertainer to receive a star on the Hollywood Walk of Fame.
- Early in her career, she recorded many popular samba songs, such as “Taí” (1930).
- Later, she appeared in many Hollywood films, such as *The Gang’s All Here* (1943).

Which choice most effectively uses information from the given sentences to begin a narrative about Carmen Miranda's life?

- A) Carmen Miranda's story begins in Marco de Canaveses, Portugal, in 1909.
- B) Carmen Miranda recorded many songs, including "Taí" (1930) and appeared in films such as *The Gang's All Here* (1943), and was the first Latin American entertainer to receive a star on the Hollywood Walk of Fame.
- C) In 1943, Carmen Miranda appeared in the Hollywood film *The Gang's All Here*.
- D) Though she would go on to achieve fame in Hollywood, Carmen Miranda began her entertainment career by recording popular samba songs, such as "Taí" (1930).

Question 27 Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

- R. Carlos Nakai is a Native American flutist and member of the Navajo-Ute nations.
- He sought to blend traditional Native American flute music with contemporary jazz and orchestral arrangements.
- Nakai's album *Changes* incorporates a traditional cedar flute, a signature instrument of many Southwestern tribes.
- Ethnomusicologist David McAllester writes that Nakai's music "draws on Native flute traditions, melodies, modes,...and other musical elements."

The student wants to connect the quotation from McAllester to a specific album. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) McAllester notes that Nakai's music "draws on Native flute traditions, melodies, modes,...and other musical elements," further indicating that it blends traditional flute music with contemporary arrangements.
- B) Consistent with McAllester's observation that Nakai's music "draws on Native flute traditions, melodies, modes,...and other musical elements," *Changes* incorporates a traditional cedar flute.
- C) McAllester's writing discusses Nakai, the Native American flutist responsible for *Changes*.
- D) Discussing Nakai's body of work, McAllester observes that it draws on elements of various Native musical traditions.

Section 1: Reading and Writing

Module 2 - Harder Path (27 Questions)

Time: 32 minutes

Question 1 *Skill: Words in Context | Difficulty: Hard*

Literary historian Kwame Asante argues that reducing the rich oral traditions of West African griots to the label “folklore” is fundamentally ____: griot performances encompass legal testimony, genealogical record-keeping, and diplomatic negotiation—functions that resist easy categorization under any single Western literary term.

Which choice completes the text with the most logical and precise word or phrase?

- A) arbitrary
 - B) reductive
 - C) indulgent
 - D) prescient
-

Question 2 *Skill: Words in Context | Difficulty: Hard*

While some critics have dismissed architect Zaha Hadid’s curvilinear designs as ____, her supporters contend that each structure’s flowing geometry is carefully calibrated to serve its intended function, from channeling foot traffic in train stations to maximizing natural light in museums.

Which choice completes the text with the most logical and precise word or phrase?

- A) meticulous
- B) derivative

- C) capricious
 - D) austere
-

Question 3 Skill: Words in Context | Difficulty: Hard

Cognitive scientist Yuna Park has demonstrated that sleep deprivation does not merely impair recall—it actively _____ memory formation, disrupting the hippocampal processes that consolidate short-term experiences into long-term neural traces.

Which choice completes the text with the most logical and precise word or phrase?

- A) precludes
 - B) augments
 - C) expedites
 - D) conflates
-

Question 4 Skill: Words in Context | Difficulty: Hard

Economist Raúl Delgado Wise has argued that trade agreements between industrialized and developing nations often _____ existing inequalities rather than ameliorate them, as the terms of such agreements tend to benefit the economies that already enjoy comparative advantages in technology and capital.

Which choice completes the text with the most logical and precise word or phrase?

- A) illuminate
 - B) circumvent
 - C) reconcile
 - D) entrench
-

Question 5 Skill: Overall Structure | Difficulty: Hard

The fruit fly *Drosophila melanogaster* has long been the organism of choice for geneticists studying hereditary mechanisms. Its short reproductive cycle (roughly ten days), large number of offspring per mating, and well-mapped genome make it ideal for controlled experiments. Yet these very qualities that make *Drosophila* useful in the laboratory may limit the generalizability of findings derived from it: its rapid generational turnover, for instance, means that evolutionary pressures observed in *Drosophila* populations may not accurately model those experienced by organisms with longer lifespans and lower reproductive rates.

Which choice best describes the overall structure of the text?

- A) It identifies an organism's traits that make it useful for research, then explains how those same traits may introduce a limitation.
- B) It contrasts two competing views about the value of a particular organism in genetic research.
- C) It presents a common misconception about the fruit fly and then provides evidence that refutes it.
- D) It describes the history of an organism's use in scientific studies and evaluates the significance of the resulting discoveries.

Question 6 Skill: Function (Underlined Portion) | Difficulty: Hard

In 1969, microbiologist Thomas D. Brock discovered *Thermus aquaticus*, a bacterium thriving in the near-boiling waters of Yellowstone's hot springs. At the time, the discovery attracted modest attention—extremophile research was a niche specialty. It was not until 1983, when biochemist Kary Mullis used an enzyme from *T. aquaticus* to develop the polymerase chain reaction (PCR), that the organism's importance became clear. PCR revolutionized molecular biology by enabling the rapid amplification of DNA segments, a technique now indispensable in forensics, medical diagnostics, and genomic research.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A) It explains why Brock's discovery was initially considered insignificant by the broader scientific community.
- B) It identifies the development that revealed the broader significance of Brock's earlier discovery.
- C) It contrasts two competing applications of extremophile research in molecular biology.
- D) It provides biographical context for Mullis's interest in extremophile organisms.

Question 7 Skill: Main Purpose | Difficulty: Hard

Anthropologist James C. Scott has influentially argued that the shift from nomadic foraging to settled agriculture was not, as traditionally assumed, a voluntary embrace of progress but rather a coerced transition driven by emerging states seeking to tax and conscript sedentary populations. Bioarchaeological evidence supports this reappraisal: skeletal remains from early agricultural communities show increased rates of dental caries, iron deficiency, and spinal degeneration compared to contemporaneous foraging populations. Yet economic historian Stefano Ferraro cautions that Scott's framework, while illuminat-

ing, may overstate the role of state coercion by underemphasizing the agency of early farming communities that adopted agriculture in regions where no centralized state apparatus yet existed.

Which choice best states the main purpose of the text?

- A) To present a scholarly argument about the origins of agriculture and introduce a qualification of that argument
- B) To contrast two incompatible theories about the health effects of early agricultural practices
- C) To argue that bioarchaeological evidence definitively resolves a longstanding debate about the shift to agriculture
- D) To explain why economic historians have rejected the conclusions drawn by anthropologists studying early agriculture

Question 8 Skill: Cross-Text | Difficulty: Hard

Text 1

Linguist Noam Chomsky has long maintained that the capacity for language is an innate biological endowment—a “universal grammar” hardwired into the human brain. Under this view, the structural similarities observed across the world’s languages reflect not cultural diffusion but shared neural architecture. Children acquire language so rapidly and with so little explicit instruction, Chomsky argues, because the deep grammatical framework is already in place at birth.

Text 2

Developmental psychologist Michael Tomasello challenges the nativist account of language acquisition by demonstrating that children learn linguistic structures through usage-based mechanisms—pattern recognition, analogy, and social interaction—rather than by activating a pre-installed grammar module. In longitudinal studies, Tomasello has shown that children’s early utterances are item-specific and formulaic, becoming more abstract only gradually as exposure to diverse sentence patterns accumulates, a trajectory inconsistent with the sudden activation of an innate system.

Based on the texts, how would Tomasello most likely respond to Chomsky’s claim that children acquire language rapidly because a deep grammatical framework is already in place at birth?

- A) By arguing that the speed of language acquisition has been exaggerated in studies that support the nativist position
- B) By contending that the gradual, pattern-driven nature of children’s early language use suggests that grammatical knowledge is constructed through experience rather than innately specified

- C) By acknowledging that an innate framework exists but maintaining that it plays a less significant role than social interaction does
- D) By proposing that language acquisition speed varies too widely across cultures to support any single explanatory model

Question 9 Skill: Main Idea | Difficulty: Hard

Recent computational analyses of large corpora of nineteenth-century periodicals have revealed that the boundary between “literary” and “popular” fiction in the Victorian era was far more permeable than traditional literary histories suggest. Scholars using topic-modeling algorithms have shown that canonical authors such as George Eliot and Charles Dickens shared thematic and stylistic features with now-forgotten writers of sensational and domestic fiction published in the same magazines. These findings do not diminish the achievements of canonical authors but instead suggest that their works emerged from—and participated in—a broader ecosystem of narrative conventions that circulated through the periodical press.

Which choice best states the main idea of the text?

- A) Computational methods have revealed that the distinction between literary and popular Victorian fiction was less rigid than scholars have traditionally maintained.
- B) Topic-modeling algorithms have proven that the works of George Eliot and Charles Dickens were primarily influenced by the conventions of sensational fiction.
- C) Scholars have recently argued that canonical Victorian authors should be considered less innovative than they were previously thought to be.
- D) Nineteenth-century periodicals published a wider variety of fiction genres than modern literary magazines do.

Question 10 Skill: Evidence - Data / Graph / Table | Difficulty: Hard

GRAPH TO INSERT

[Table: Mean Annual Precipitation and Primary Vegetation Type at Four Research Stations in East Africa]

Station	Elevation (m)	Mean annual precipitation (mm)	Primary vegetation
Marsabit	1,345	254	Dry scrubland
Nanyuki	1,947	680	Montane grassland
Nyeri	1,759	988	Mixed forest

Station	Elevation (m)	Mean annual precipitation (mm)	Primary vegetation
Kericho	2,004	1,512	Dense tropical forest

An ecologist studying the relationship between precipitation and vegetation in East Africa hypothesizes that mean annual precipitation, rather than elevation alone, is the primary determinant of vegetation type across these stations.

Which choice best describes data from the table that support the ecologist's hypothesis?

- A) Marsabit, the station with the lowest elevation, also has the least precipitation, which suggests that elevation and precipitation are interchangeable predictors of vegetation type.
- B) Although Nyeri has a lower elevation than Nanyuki, Nyeri receives more precipitation and supports denser vegetation, consistent with precipitation being the key factor.
- C) Kericho has both the highest elevation and the highest precipitation among the four stations, suggesting that elevation drives precipitation and therefore vegetation type.
- D) Nanyuki and Nyeri differ in elevation by only 188 meters yet support similar vegetation types, which indicates that neither precipitation nor elevation influences vegetation.

Question 11 Skill: Evidence - Support / Weaken | Difficulty: Hard

Geochemist Elif Yılmaz and colleagues analyzed sediment cores from Lake Van in eastern Turkey to reconstruct volcanic activity over the past 600,000 years. They identified over 250 distinct tephra layers—deposits of volcanic ash—and correlated each layer with known eruptions using radiometric dating. The team concluded that the frequency of major eruptions in the region has not been constant but instead shows pronounced clustering, with periods of intense activity separated by long quiescent intervals.

Which finding, if true, would most directly support the team's conclusion?

- A) The chemical composition of the tephra layers indicates that most eruptions originated from the same volcanic source.
- B) The distribution of dated tephra layers shows statistically significant groupings at intervals of approximately 40,000 to 80,000 years, with intervening gaps containing few or no layers.

- C) Lake Van's sedimentary record extends further back in time than that of any other lake in the region.
- D) Tephra layers from the most recent 100,000 years are better preserved than those from earlier periods.

Question 12 Skill: Evidence - Support / Weaken | Difficulty: Hard

Marine biologists Amara Osei and Luis Herrera have observed that certain deep-sea anglerfish species maintain bioluminescent lures whose light intensity varies with depth. They hypothesize that this variation is an adaptive response to ambient light conditions: at moderate depths, where some sunlight penetrates, a dimmer lure avoids detection by the anglerfish's own predators, while at extreme depths, where no ambient light exists, a brighter lure is more effective at attracting prey without incurring additional predation risk.

Which finding, if true, would most directly support Osei and Herrera's hypothesis?

- A) Anglerfish species at extreme depths produce light using different chemical reactions than do species at moderate depths.
- B) The brightest lures observed belong to species living at depths where ambient light levels are lowest, and species at moderate depths produce lures too dim to be visible to predators situated above them.
- C) Anglerfish at all depths exhibit similar metabolic rates, suggesting that lure intensity is not constrained by energy availability.
- D) Deep-sea prey organisms are attracted to a wider range of light intensities than previously believed.

Question 13 Skill: Evidence - Support / Weaken | Difficulty: Hard

Neuroscientists Javier Moreno and Priya Chakraborty studied how bilingual individuals switch between languages during conversation. They monitored brain activity in 40 Spanish-English bilinguals during a task that required alternating between languages every few sentences. The researchers concluded that the cognitive cost of language switching diminishes with bilingual proficiency: highly proficient bilinguals show less neural activation in executive control regions during switching than do less proficient bilinguals.

Which finding from the study, if true, would most directly support the researchers' conclusion?

- A) Bilinguals who reported using both languages daily showed slower reaction times during the switching task than did those who used only one language at work.

- B) The difference in prefrontal cortex activation between switching and non-switching trials was smaller for participants who scored in the top quartile on a bilingual proficiency assessment than for those in the bottom quartile.
 - C) All participants, regardless of proficiency, showed increased activation in auditory processing regions when hearing the less-dominant language.
 - D) Participants who had learned their second language before age five showed greater overall brain volume than those who learned it later.
-

Question 14 Skill: Quotation Illustrates Claim | Difficulty: Hard

North and South is an 1854 novel by Elizabeth Gaskell. In the novel, the narrator suggests that the protagonist, Margaret Hale, experiences an internal conflict between her sense of moral duty and her instinct for self-preservation: _____

Which quotation from *North and South* most effectively illustrates the claim?

- A) "Margaret felt that she could not bear to be spoken to just then; the only motion she made was to put her hands together, pressing the fingers tight."
 - B) "She tried to settle to her work, but an unseen restlessness possessed her, and she wandered from room to room."
 - C) "Margaret's conscience told her to speak, even as every nerve in her body urged her to stay silent, to let the moment pass safely by."
 - D) "It was one of those actions that, once done, could not be recalled; yet she would not have it otherwise."
-

Question 15 Skill: Inference / Complete the Text | Difficulty: Hard

The prevailing model of coral bleaching attributes the phenomenon primarily to thermal stress: elevated sea temperatures cause corals to expel their symbiotic algae, resulting in the characteristic whitening. Marine ecologist Sofia Andersen, however, has documented multiple instances of bleaching in reef systems where water temperatures remained within the historical range for the region. Andersen's observations suggest that _____

Which choice most logically completes the text?

- A) coral bleaching may be driven by factors beyond thermal stress alone.

- B) the prevailing model of coral bleaching overstates the role of symbiotic algae in maintaining coral health.
 - C) coral species that have already undergone bleaching are more resistant to future episodes of thermal stress.
 - D) reefs in regions with historically stable water temperatures are unlikely to experience bleaching under any conditions.
-

Question 16 Skill: Inference / Complete the Text | Difficulty: Hard

Historians of science have typically attributed the rapid adoption of germ theory in the late nineteenth century to the persuasive experimental demonstrations of Louis Pasteur and Robert Koch. Sociologist of science Bruno Latour, however, has argued that the acceptance of germ theory depended less on the intrinsic strength of the experimental evidence than on Pasteur's ability to recruit powerful allies—physicians, public health officials, and agricultural interests—who had their own pragmatic reasons for endorsing the new framework. If Latour's account is correct, then the acceptance of germ theory is best understood as _____

Which choice most logically completes the text?

- A) a political and institutional achievement as much as a purely scientific one.
 - B) a demonstration that experimental evidence is unnecessary for scientific progress.
 - C) an anomaly in the history of science that cannot be replicated under modern conditions.
 - D) evidence that Pasteur's experiments were less rigorous than Koch's.
-

Question 17 Skill: Grammar / Conventions | Difficulty: Hard
ture — Modifier Placement)

Engineered to withstand sustained temperatures exceeding 1,600 degrees Celsius, the new generation of ceramic thermal shields unveiled by aerospace manufacturer Meridian Dynamics _____ crewed spacecraft during atmospheric reentry.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) are designed to protect
- B) designing the protection of
- C) having been designed protecting
- D) and are designed to protect

Question 18 Skill: Grammar / Conventions | Difficulty: Hard
Semicolon vs. Colon)

The excavation at Göbekli Tepe upended a central assumption of archaeology ____ complex monumental construction, scholars had long believed, required the social organization that only settled agricultural communities could provide.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) ;
- B) :
- C) ,
- D) —and

Question 19 Skill: Grammar / Conventions | Difficulty: Hard
Agreement — Complex Subject)

Neither the rising cost of raw materials nor the implementation of new import tariffs on finished goods ____ entirely responsible for the sharp decline in domestic furniture manufacturing observed between 2015 and 2020.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) are
- B) were
- C) is
- D) have been

Question 20 Skill: Grammar / Conventions | Difficulty: Hard
sequence)

By the time astronomer Vera Rubin published her landmark 1980 paper on galaxy rotation curves, she ____ nearly two decades collecting observational data that would ultimately provide compelling evidence for the existence of dark matter.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) has spent
- B) had spent
- C) was spending
- D) would spend

Question 21 Skill: Grammar / Conventions | Difficulty: Hard
(*strictive Clause*)

The composer whose symphonies _____ most frequently performed during the 2023 Berlin Philharmonic season was Ludwig van Beethoven, accounting for twelve of the orchestra's forty-eight scheduled concerts.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) were
- B) , were
- C) were,
- D) , were,

Question 22 Skill: Grammar / Conventions | Difficulty: Hard

The museum's new conservation plan calls for digitizing all archival photographs, _____ the humidity levels in each storage vault, and retraining staff in artifact-handling protocols.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) to regulate
- B) regulation of
- C) they will regulate
- D) regulating

Question 23 Skill: Transitions | Difficulty: Hard

The construction of the Hoover Dam in the 1930s required the diversion of the entire Colorado River through four massive tunnels blasted into the canyon walls. _____ the dam's concrete had to be poured in small, individually cooled blocks rather than as a single mass, because engineers calculated that a monolithic pour would have taken over a century to cool and would have cracked catastrophically in the process.

Which choice completes the text with the most logical transition?

- A) Similarly,
- B) Moreover,
- C) In other words,
- D) Nevertheless,

Question 24 Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

- The kākākāpō (*Strigops habroptilus*) is a flightless, nocturnal parrot native to New Zealand.
- By 1995, the wild population had declined to 51 individuals.
- New Zealand's Department of Conservation launched an intensive breeding and relocation program.
- By 2024, the population had grown to approximately 250 individuals.
- The species remains critically endangered due to its low genetic diversity and vulnerability to introduced predators.

The student wants to emphasize the contrast between the conservation program's achievements and the species' ongoing vulnerability. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The kākākāpō is a flightless parrot native to New Zealand whose population declined to 51 individuals by 1995.
- B) Thanks to an intensive breeding program by New Zealand's Department of Conservation, the kākākāpō population grew from 51 in 1995 to approximately 250 by 2024.
- C) Although a conservation program has increased the kākākāpō population from 51 individuals in 1995 to roughly 250 by 2024, the species remains critically endangered due to low genetic diversity and vulnerability to introduced predators.
- D) New Zealand's Department of Conservation launched a breeding program for the kākākāpō, a flightless, nocturnal parrot that is critically endangered.

Question 25 Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

- The Dead Sea is a hypersaline lake bordered by Jordan, Israel, and the West Bank.
- Its surface is approximately 430 meters below sea level, making it the lowest land elevation on Earth.
- Its salinity is approximately 34%, roughly ten times that of the open ocean.
- The high salinity prevents macroscopic aquatic organisms such as fish from surviving in the water.
- The lake supports populations of halophilic (salt-loving) microorganisms, including archaea and certain bacteria.

The student wants to explain why the Dead Sea supports some life but not others. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The Dead Sea, the lowest land elevation on Earth, has a salinity of approximately 34%, which is roughly ten times that of the open ocean.
- B) Bordered by Jordan, Israel, and the West Bank, the Dead Sea is a hypersaline lake whose surface lies approximately 430 meters below sea level.
- C) The Dead Sea's salinity of approximately 34% is lethal to fish and other macroscopic aquatic organisms, yet halophilic archaea and bacteria thrive in the water because they are adapted to extreme salt concentrations.
- D) The Dead Sea supports populations of halophilic microorganisms, including archaea and certain bacteria, in addition to various species of fish.

Question 26 *Skill: Rhetorical Synthesis | Difficulty: Hard*

While researching a topic, a student has taken the following notes:

- Architect I. M. Pei designed the East Building of the National Gallery of Art in Washington, D.C.
- The building opened in 1978.
- Its design features sharp geometric forms, including a prominent triangular floor plan.
- Pei also designed the glass and metal pyramid at the Louvre Museum in Paris.
- The Louvre Pyramid was completed in 1989.
- Both structures use geometric simplicity to complement their surrounding historic architecture.

The student wants to draw a connection between the two projects. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) I. M. Pei's East Building of the National Gallery of Art, which opened in 1978, features a prominent triangular floor plan.
- B) Pei's glass and metal pyramid at the Louvre Museum in Paris was completed in 1989, more than a decade after the East Building opened.
- C) In both the East Building of the National Gallery (1978) and the Louvre Pyramid (1989), Pei used geometric simplicity to create modern structures that complement their surrounding historic architecture.

- D) The Louvre Pyramid, completed in 1989, is one of the most recognizable landmarks in Paris.
-

Question 27 *Skill: Rhetorical Synthesis | Difficulty: Hard*

While researching a topic, a student has taken the following notes:

- The Great Barrier Reef is the world's largest coral reef system, stretching over 2,300 kilometers along Australia's northeastern coast.
- Mass bleaching events were recorded in 1998, 2002, 2016, 2017, 2020, 2022, and 2024.
- The interval between mass bleaching events has shortened over the past three decades.
- Scientists attribute the increasing frequency to rising ocean temperatures driven by climate change.
- Shorter intervals between bleaching events reduce the time available for coral recovery.

The student wants to emphasize the trend in bleaching frequency and its implications for coral recovery. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The Great Barrier Reef, the world's largest coral reef system, has experienced mass bleaching events in multiple years, including 1998, 2016, and 2024.
- B) As ocean temperatures rise due to climate change, the Great Barrier Reef has experienced mass bleaching events at increasingly short intervals, leaving less time between episodes for damaged coral to recover.
- C) Scientists attribute mass bleaching events on the Great Barrier Reef to rising ocean temperatures, a consequence of climate change.
- D) The Great Barrier Reef stretches over 2,300 kilometers along Australia's northeastern coast and is the largest coral reef system in the world.
-

Section 2: Math

Module 1 (22 Questions) - All Students

Time: 35 minutes

Question 1 Skill: Lines & Angles | Difficulty: Medium

[Figure: Line m intersects lines p and q . Line p is parallel to line q . The angle formed between line m and line p on the upper-right side is labeled 144° . The angle formed between line m and line q on the lower-left side is labeled y° .]

Note: Figure not drawn to scale.

In the figure, line m intersects lines p and q . Line p is parallel to line q . What is the value of y ?

Question 2 Skill: Linear Inequalities | Difficulty: Medium

During a section of a race, a competitive cyclist's speed varied between 22 kilometers per hour and 28 kilometers per hour. Which inequality best represents this situation, where v is the speed, in kilometers per hour, during this section of the race?

- A) $v \leq 22$
 - B) $v \leq 28$
 - C) $22 \leq v$
 - D) $22 \leq v \leq 28$
-

Question 3 Skill: Linear Equations in One & Two Variables | Difficulty: Medium

If $7x = 3$, what is the value of $56x$?

- A) 3
 - B) 8
 - C) 21
 - D) 24
-

Question 4 Skill: Ratios, Rates & Proportions | Difficulty: Medium

Which expression is equivalent to $(x^3 + 6x^2 - 5x) + 3(x^2 + 12)$?

- A) $x^3 + 9x^2 - 5x + 36$
 - B) $x^3 + 6x^2 - 5x + 36$
 - C) $x^3 + 9x^2 - 5x + 12$
 - D) $x^3 + 7x^2 - 5x + 36$
-

Question 5 Skill: Function Notation & Evaluation | Difficulty: Medium

The function f is defined by $f(x) = x^3 + 5$. What is the value of $f(3)$?

Question 6 Skill: Linear Equations in One & Two Variables | Difficulty: Medium

The function g is defined by $g(x) = x/4$. For what value of x does $g(x) = 318$?

Question 7 Skill: Linear & Exponential Models from Data | Difficulty: Medium

[Scatterplot: Shows the relationship between two variables, x and y . Points are approximately at: (1, 9), (2, 8), (2, 7), (3, 6), (3, 6), (4, 5), (5, 5), (6, 4), (7, 4), (8, 3), (9, 2), (10, 1). The trend is clearly negative and approximately linear.]

Which of the following equations is the most appropriate linear model for the data shown?

- A) $y = 1.2 + 9.5x$
 - B) $y = 1.2 - 9.5x$
 - C) $y = 9.5 + 1.2x$
 - D) $y = 9.5 - 0.9x$
-

Question 8 Skill: Quadratic Functions - Graphs & Properties | Difficulty: Medium

The function $f(x) = (1/16)(x - 6)^2 + 3$ gives a toy rocket's height above the ground $f(x)$, in feet, x seconds after it started ascending on a launch pad, where $0 \leq x \leq 12$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy -plane?

- A) The toy rocket's minimum height was 3 feet above the ground.
 - B) The toy rocket's minimum height was 6 feet above the ground.
 - C) The toy rocket's height was 3 feet above the ground when it started moving.
 - D) The toy rocket's height was 6 feet above the ground when it started moving.
-

Question 9 Skill: Linear Equations in One & Two Variables | Difficulty: Medium

$$5x - 3y = -45$$

For the given equation, which table gives three values of x and their corresponding values of y ?

x	0	3	6
y	15	20	25

A)

x	0	3	6
y	25	20	15

B)

x	15	20	25
y	0	3	6

C)

x	15	20	25
y	6	3	0

D)

Question 10 Skill: Creating Linear Models from Word Problems | Difficulty: Medium

A pharmacist combines saline solution and medication to make a mixture with a total volume of 60 milliliters (mL). The volume of medication in the mixture is 15 mL. What is the volume of saline solution, in mL, in the mixture? (Assume that the volume of the mixture is the sum of the volumes of saline solution and medication before they were mixed.)

Question 11 Skill: Systems of Linear Equations | Difficulty: Medium

$$y = -(1/5)x \quad y = (1/13)x$$

The solution to the given system of equations is (x, y). What is the value of x?

- A) -5
- B) 0
- C) 5

D) 13

Question 12 Skill: Exponential Functions - Growth & Decay | Difficulty: Medium

A city has a bicycle-sharing program. In March 2020, there were 2,400 active subscribers to the program. For the next 18 months after March 2020, the total number of active subscribers each month was 8% greater than the total number the previous month. Which equation gives the total number of subscribers, s , enrolled in the bicycle-sharing program m months after March 2020, where $m \leq 18$?

- A) $s = 2,400(0.08)^m$
 - B) $s = 2,400(8)^m$
 - C) $s = 2,400(1.8)^m$
 - D) $s = 2,400(1.08)^m$
-

Question 13 Skill: Exponential Functions - Growth & Decay | Difficulty: Medium

The function $f(t) = 5,000(2)^{t/40}$ gives the number of cells in a culture t minutes after an initial observation. How much time, in minutes, does it take for the number of cells in the culture to double?

Question 14 Skill: Systems of Equations - Number of Solutions | Difficulty: Medium

$$y = (3/8)x + 7$$

One of two equations in a system of linear equations is given. The system has infinitely many solutions. What is the slope of the graph of the second equation?

- A) $-8/3$
 - B) $-3/8$
 - C) $3/8$
 - D) $8/3$
-

Question 15 Skill: Quadratic Equations - Solving | Difficulty: Medium

If $6x^2 - 18x - 36 = 0$, what is the value of $x^2 - 3x$?

- A) 0
 - B) 3
 - C) 4
 - D) 6
-

Question 16 Skill: Volume & Surface Area | Difficulty: Medium

A rectangular prism has a volume of 1,728 cubic centimeters and all edges of equal length (i.e., it is a cube). What is the surface area, in square centimeters, of the cube?

Question 17 Skill: Creating Linear Models from Word Problems | Difficulty: Medium

Priya bought 7 notebooks that were each the same price. She used a coupon for \$35 off the entire purchase. The cost for the entire purchase after using the coupon was \$91. What was the original price, in dollars, for 1 notebook?

Question 18 Skill: Discriminant & Number of Solutions | Difficulty: Medium

$$-3x^2 + bx - 20 = 0$$

In the given equation, b is a positive integer. The equation has no real solution. What is the largest possible value of b ?

Question 19 Skill: Right Triangle Trigonometry | Difficulty: Medium

In right triangle DEF, angles D and E are acute, side DF has a length of 18.6, and $\tan E = 1/5$. What is the length of side EF, rounded to the nearest tenth?

- A) 3.7
 - B) 5.2
 - C) 93.0
 - D) 379.1
-

Question 20 Skill: Quadratic Functions - Graphs & Properties | Difficulty: Medium

$$f(x) = (x - 5)^2 + 9$$

What is the minimum value of the given function?

- A) 5
 - B) 9
 - C) 14
 - D) 16
-

Question 21 Skill: Circle Equations | Difficulty: Medium

Point A lies on a unit circle in the xy -plane and has coordinates $(0, 1)$. Point B is the center of the circle and has coordinates $(0, 0)$. Point

C also lies on the circle and has coordinates $(0, -1)$. Which of the following could be the positive measure of angle ABC, in radians?

- A) $21\pi/2$
 - B) $23\pi/2$
 - C) 20π
 - D) 21π
-

Question 22 *Skill: Similar Triangles | Difficulty: Medium*

For triangle JKL and triangle MNO, angles J and M each measure 45° , $KL = 20$ and $NO = 15$. Which additional piece(s) of information is(are) sufficient to prove that triangle JKL is similar to triangle MNO?

I. The length of JK is $(4/3)$ the length of MN. II. The length of JL is $(4/3)$ the length of MO.

- A) I is sufficient, but II is not.
 - B) II is sufficient, but I is not.
 - C) I is sufficient, and II is sufficient.
 - D) Neither I nor II is sufficient.
-

Section 2: Math

Module 2 - Easier Path (22 Questions)

Time: 35 minutes

Question 1 Skill: Graphs of Linear Equations | Difficulty: Easy

$$f(x) = 6x + c$$

For the linear function f , c is a constant and $f(0) = 13$. What is the value of c ?

Question 2 Skill: Statistics: Mean & Median | Difficulty: Easy

Data set P: 3, 7, 10, 10, 13, 19 Data set Q: 3, 7, 10, 10, 13, 19, 38

Two data sets are shown. Which statement best compares the medians of the data sets?

- A) The median of data set P is greater than the median of data set Q.
 - B) The median of data set P is less than the median of data set Q.
 - C) The medians of data sets P and Q are equal.
 - D) There is not enough information to compare the medians.
-

Question 3 Skill: Statistical Inference & Sampling | Difficulty: Easy

To estimate the proportion of a population that has a certain trait, a random sample was selected from the population. Based on the sample, it is estimated that the proportion of the population that has the trait is 0.35, with an associated margin of error of 0.06. Based on this estimate and margin of error, which of the following is the most appropriate conclusion about the proportion of the population that has the trait?

- A) It is plausible that the proportion is between 0.29 and 0.41.
- B) It is plausible that the proportion is less than 0.29.
- C) The proportion is exactly 0.35.
- D) It is plausible that the proportion is greater than 0.41.

Question 4 Skill: Graphs of Linear Equations | Difficulty: Easy

[Graph: A coordinate plane showing a line passing through approximately (0, 0) and (4, 8) with slope 2. The shaded region is to the LEFT of the line (above the line, which goes from lower-left to upper-right). The line appears dashed.]

The shaded region shown represents solutions to an inequality. Which ordered pair (x, y) is a solution to this inequality?

- A) (−6, 0)
- B) (0, 6)
- C) (0, −6)
- D) (6, 0)

Question 5 Skill: Graphs of Linear Equations | Difficulty: Easy
Tables

x	f(x)
−1	6
0	10
1	18

For the quadratic function f , the table shows three values of x and their corresponding values of $f(x)$. Which equation defines f ?

- A) $f(x) = 2x^2 + 6x + 10$
- B) $f(x) = 7x^2 + x + 10$
- C) $f(x) = 7x^2 - x + 10$
- D) $f(x) = x^2 + 7x + 10$

Question 6 Skill: Exponential Functions - Growth & Decay | Difficulty: Easy

[Graph: A coordinate plane with x-axis labeled 0 to 5 and y-axis labeled 0 to 14 (in thousands). A curve starts at approximately (0, 5) and increases exponentially, passing through approximately (1, 7), (2, 9), and reaching about (3, 13) at $x = 3$.]

The graph gives the estimated revenue y , in thousands of dollars, of a startup company x years since 2019, where $0 \leq x \leq 5$. Which of the following best describes the increase in the estimated revenue from $x = 0$ to $x = 1$?

- A) The estimated revenue at $x = 1$ is 3.0 times the estimated revenue at $x = 0$.
- B) The estimated revenue at $x = 1$ is 2.0 times the estimated revenue at $x = 0$.
- C) The estimated revenue at $x = 1$ is 1.4 times the estimated revenue at $x = 0$.
- D) The estimated revenue at $x = 1$ is 0.7 times the estimated revenue at $x = 0$.

Question 7 Skill: Interpreting Coefficients in Linear Models | Difficulty: Easy

$$C(x) = 85 - 5x$$

The function C estimates the remaining charge, in percent, of a laptop battery on a certain day, where x is the number of hours after 9 a.m. and $1 \leq x \leq 8$. According to the function, what was the estimated decrease in charge per hour, in percent, during this time period?

- A) 5
- B) 9
- C) 80
- D) 85

Question 8 Skill: Special Right Triangles | Difficulty: Easy

[Figure: A right triangle with a right angle at the bottom-left vertex. The bottom side (horizontal) is labeled 48. The angle at the bottom-right vertex is labeled 30° . The left side (vertical) is labeled w .]

Note: Figure not drawn to scale.

In the triangle shown, what is the value of w ?

- A) 16
- B) $16\sqrt{3}$

- C) $48\sqrt{3}$
 - D) 96
-

Question 9 Skill: Ratios, Rates & Proportions | Difficulty: Easy

There is a proportional relationship between q and r . If $q = 3,150$ when $r = 4,200$, then what is the value of q when $r = 3,360$?

- A) 2,310
 - B) 4,480
 - C) 4,410
 - D) 2,520
-

Question 10 Skill: Volume & Surface Area | Difficulty: Easy

A cube has a volume of 64,000 cubic units. What is the surface area, in square units, of the cube?

Question 11 Skill: Systems of Equations - Number of Solutions | Difficulty: Easy

$$(6(cx + 4))/5 = 3x - 2$$

In the given equation, c is a constant. If the equation has no solution, what is the value of c ?

- A) $-5/2$
 - B) $-2/5$
 - C) $2/5$
 - D) $5/2$
-

Question 12 Skill: Interpreting Coefficients in Linear Models | Difficulty: Easy

The function $f(x)$ is defined as 21 more than 7 times a number x . If $y = f(x)$ is graphed in the xy -plane, what is the best interpretation of the x -intercept?

- A) When $f(x) = 0$, the number is $-21/7$.
 - B) When the number is 0, $f(x) = 21$.
 - C) For each increase of 1 in the value of the number, $f(x)$ increases by 7.
 - D) The value of $f(x)$ increases by 1 for each increase of 7 in the value of the number.
-

Question 13 Skill: Ratios, Rates & Proportions | Difficulty: Easy

$$3/c - 5/d = 4$$

The given equation relates the variables c and d , where c and d are greater than 1. Which equation correctly expresses d in terms of c ?

- A) $d = c / (5(3 - 4c))$
 - B) $d = 5c / (3 - 4c)$
 - C) $d = 5(3 - 4c) / c$
 - D) $d = (3 - 4c) / (5c)$
-

Question 14 Skill: Systems of Linear Equations | Difficulty: Easy

A proposal for a park renovation was included on a city ballot. A local newspaper stated that 6 times as many people voted in favor of the proposal as people who voted against it. A community blog reported that 18,000 more people voted in favor of the proposal than voted against it. Based on these data, how many people voted against the proposal?

Question 15 Skill: Function Notation & Evaluation | Difficulty: Easy

The function f is defined by $f(x) = 3x^2$. What is the value of $f(12)$?

- A) 36
 - B) 72
 - C) 108
 - D) 432
-

Question 16 Skill: Radical and Rational Exponents | Difficulty: Easy

$$\sqrt[4]{(p^3)} = t^{(5/4)}$$

In the given equation, $p > 1$ and $t > 1$. If $t = p^{(2n-1)}$, where n is a constant, what is the value of n ?

Question 17 Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Easy

The equation $6|x - 5| = k$, where k is a constant, has exactly one solution. Which of the following could be the value of $k/6$?

- A) -5 only
 - B) -5 or 5
 - C) 0 only
 - D) 5 only
-

Question 18 Skill: Discriminant & Number of Solutions | Difficulty: Easy

$$-5x^2 + bx - 18 = 0$$

In the given equation, b is a positive integer. The equation has no real solution. What is the largest possible value of b ?

Question 19 Skill: Exponential Functions - Growth & Decay | Difficulty: Easy

The function f is defined by $f(x) = a(2.5^x + 2.5^b)$, where a and b are integer constants and $0 < a < b$. The functions g and h are equivalent to function f , where k and m are constants. Which of the following equations displays the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane as a constant or coefficient?

I. $g(x) = a(2.5^x + k)$ II. $h(x) = a(2.5)^x + m$

- A) I only
 - B) II only
 - C) I and II
 - D) Neither I nor II
-

Question 20 Skill: Creating Linear Models from Word Problems | Difficulty: Easy

The cost of renting a photography studio for up to 4 hours is \$180 for the first hour and \$95 for each additional hour. Which of the following equations gives the cost y , in dollars, of renting the studio for x hours, where x is a positive integer and $x \leq 4$?

- A) $y = 180x + 95$
 - B) $y = 180x - 95$
 - C) $y = 95x + 180$
 - D) $y = 95x + 85$
-

Question 21 Skill: Percentages | Difficulty: Easy

The mass of object X is 375% of the mass of object Y, and the mass of object X is 0.625% of the mass of object Z. If the mass of object Z is $q\%$ of the mass of object Y, what is the value of q ?

Question 22 Skill: Similar Triangles | Difficulty: Easy

For triangle ABC and triangle DEF, angles A and D each measure 40° , $BC = 18$ and $EF = 12$. Which additional piece(s) of information is(are) sufficient to prove that triangle ABC is similar to triangle DEF?

I. The length of AB is $(3/2)$ the length of DE. II. The length of AC is $(3/2)$ the length of DF.

- A) I is sufficient, but II is not.
- B) II is sufficient, but I is not.
- C) I is sufficient, and II is sufficient.
- D) Neither I nor II is sufficient.

Section 2: Math

Module 2 - Harder Path (22 Questions)

Time: 35 minutes

Question 1 *Skill: Nonlinear Systems of Equations | Difficulty: Hard*

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$$y = x^2 - 6x + 5 \quad y = 2x - 7$$

How many solutions does the given system of equations have?

- A) 0
 - B) 1
 - C) 2
 - D) Infinitely many
-

Question 2 Skill: Ratios, Rates & Proportions | Difficulty: Hard

Which expression is equivalent to $(2x^2 + 9x + 9) / (x^2 - 9)$ for all values of x where the expression is defined?

- A) $(2x - 3) / (x - 3)$
 - B) $(2x - 3) / (x + 3)$
 - C) $(2x + 3) / (x - 3)$
 - D) $(2x + 3) / (x + 3)$
-

Question 3 Skill: Quadratic Equations - Solving | Difficulty: Hard

$$x^2 - kx + 36 = 0$$

In the given equation, k is a positive integer. For how many distinct positive integer values of k does the equation have two positive integer solutions?

- A) 3
 - B) 4
 - C) 5
 - D) 6
-

Question 4 Skill: Function Notation & Evaluation | Difficulty: Hard

The functions f and g are defined by $f(x) = 3x - 2$ and $g(x) = x^2 + 1$. What is the value of $f(g(4))$?

- A) 11
 - B) 37
 - C) 49
 - D) 51
-

Question 5 Skill: Exponential Equations & Modeling | Difficulty: Hard

If $3^{(2x+1)} = 243$, what is the value of x ?

Question 6 Skill: Factoring Polynomials | Difficulty: Hard

$$p(x) = x^3 - 2x^2 - 13x - 10$$

It can be shown that $(x + 1)$ is a factor of $p(x)$. Which of the following correctly expresses $p(x)$ as the product of linear factors?

- A) $(x + 1)(x - 5)(x + 2)$
 - B) $(x + 1)(x + 5)(x - 2)$
 - C) $(x + 1)(x - 5)(x - 2)$
 - D) $(x - 1)(x + 5)(x + 2)$
-

Question 7 Skill: Circle Equations | Difficulty: Hard

A circle in the xy -plane has equation $x^2 + y^2 - 8x + 6y = 24$. What is the radius of the circle?

- A) 5
 - B) 7
 - C) 24
 - D) 49
-

Question 8 Skill: Circle Equations | Difficulty: Hard

If $\sin \theta = 5/13$ and θ is in the second quadrant ($\pi/2 < \theta < \pi$), what is the value of $\cos \theta$?

- A) $-12/13$
 - B) $12/13$
 - C) $-5/13$
 - D) $5/12$
-

Question 9 Skill: Similar Triangles | Difficulty: Hard
soning

[Figure: In triangle ABC, point D lies on side AB and point E lies on side AC such that DE is parallel to BC. AD = 6, DB = 9, and DE = 8.]

Note: Figure not drawn to scale.

In the figure, DE is parallel to BC in triangle ABC. AD = 6, DB = 9, and DE = 8. What is the length of BC?

- A) 12
 - B) 16
 - C) 18
 - D) 20
-

Question 10 Skill: Standard Deviation | Difficulty: Hard

Which of the following data sets has the greatest standard deviation?

- A) {10, 10, 10, 10, 10}
 - B) {8, 9, 10, 11, 12}
 - C) {2, 6, 10, 14, 18}
 - D) {10, 10, 10, 10, 50}
-

Question 11 Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Hard

What is the minimum value of the expression $2x^2 - 12x + 23$?

- A) 3
 - B) 5
 - C) 7
 - D) 23
-

Question 12 Skill: Systems of Linear Equations | Difficulty: Hard

A florist is making bouquets using roses and lilies. Each bouquet must contain exactly 12 flowers. Roses cost \$3 each and lilies cost \$5 each. If the florist spends exactly \$44 on flowers for one bouquet, how many roses are in the bouquet?

- A) 4
 - B) 7
 - C) 8
 - D) 9
-

Question 13 Skill: Ratios, Rates & Proportions | Difficulty: Hard

If $(x^{(2/3)})^6 = x^n$, what is the value of n ?

Question 14 Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Hard

If $a - b = 5$ and $a^2 - b^2 = 40$, what is the value of $a + b$?

Question 15 Skill: Linear Inequalities | Difficulty: Hard

For what values of x is $x^2 - 5x - 6 < 0$?

- A) $x < -1$ or $x > 6$
- B) $-1 < x < 6$
- C) $x < -6$ or $x > 1$
- D) $-6 < x < 1$

Question 16 Skill: Circle Equations | Difficulty: Hard

A circle has a radius of 10 centimeters. What is the length, in centimeters, of an arc subtended by a central angle of $3\pi/5$ radians?

- A) 2π
 - B) 5π
 - C) 6π
 - D) $50\pi/3$
-

Question 17 Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

An investment of \$2,000 earns interest compounded annually at a rate of 6% per year. Which expression gives the value of the investment, in dollars, after t years?

- A) $2,000(0.06)^t$
 - B) $2,000(1.06)^t$
 - C) $2,000 + 120t$
 - D) $2,000(6)^t$
-

Question 18 Skill: Systems of Linear Equations | Difficulty: Hard

A bakery sells muffins for \$3 each and cookies for \$2 each. On Monday, the bakery sold a total of 50 muffins and cookies combined and collected \$130 in revenue from those items. How many muffins did the bakery sell on Monday?

- A) 20
 - B) 25
 - C) 30
 - D) 35
-

Question 19 Skill: Quadratic Functions - Graphs & Properties | Difficulty: Hard
tions

GRAPH TO INSERT

If r and s are solutions to $3x^2 - 15x + 11 = 0$, what is the value of $(1/r) + (1/s)$?

- A) $11/15$
 - B) $11/3$
 - C) $3/11$
 - D) $15/11$
-

Question 20 Skill: Volume & Surface Area | Difficulty: Hard

A right circular cone has a height of 12 inches and a base radius of 5 inches. What is the volume, in cubic inches, of the cone?

- A) 60π
 - B) 100π
 - C) 300π
 - D) $100\pi/3$
-

Question 21 Skill: Percentages | Difficulty: Hard

The price of a stock increased by 20% in January and then decreased by 25% in February. What was the net percent change in the stock price over the two months?

- A) -10%
 - B) -5%
 - C) 0%
 - D) 5%
-

Question 22 Skill: Parallel & Perpendicular Lines | Difficulty: Hard

Line m passes through the points (1, 4) and (3, 10). Line n is perpendicular to line m and passes through the point (6, 2). What is the y -intercept of line n ?

- A) 1
 - B) 3
 - C) 4
 - D) 5
-

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